

SARRA MISSAOUI

Software Engineer — Graduate Studies Applicant

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PROFILE

Software engineering graduate (2025) with hands-on experience building full-stack web and mobile applications across internships and independent projects. Skilled in Java/Spring Boot, JavaScript frameworks, and cloud/DevOps tooling, with a growing interest in AI/ML systems. Seeking admission to a Master's program to deepen expertise in software engineering and applied artificial intelligence.

EDUCATION

Software Engineering Degree — Polytechnique Sousse 2022 – 2025

Integrated Preparatory Cycle — Polytechnique Sousse 2020 – 2022

Baccalaureate, Experimental Sciences — Hammam Sousse 2 High School 2020

PROFESSIONAL EXPERIENCE

Final-Year Intern, ISIS Institute Feb 2025 – May 2025

Abidjan

- Designed and developed a full-stack web application to digitalize academic management for an educational institution, centralizing student, teacher, course, and grade data.
- Built RESTful APIs with Spring Boot, Spring Security, and JPA/Hibernate; developed a responsive front end with React.js.
- Automated administrative workflows and delivered role-based interfaces that improved communication and transparency across the institution.

Technologies: Spring Boot, Spring Security, JPA/Hibernate, RESTful APIs, React.js, HTML5, CSS3, JavaScript (ES6), MySQL, Git, Maven

Summer Intern, VegaNet Jul 2024 – Sep 2024

Sousse, Tunisia

- Designed and implemented a requirements-management module to collect, track, and prioritize client requests.
- Developed a contract-management module in Odoo to automate contract creation, tracking, and renewal.

Technologies: Odoo, Python, JavaScript

Summer Intern, ISIS Institute Jul 2023 – Aug 2023

Sousse, Tunisia — Subject: Student Enrollment & Academic Records Management

- Designed and developed a web module for managing student enrollment, academic records, and grade tracking within the institution's information system.
- Implemented role-based access for administrators and staff to streamline data entry, consultation, and reporting.
- Contributed to improving the reliability, consistency, and accessibility of academic data across departments.

Technologies: PHP, Symfony, MySQL, HTML5, CSS3

SELECTED ACADEMIC & PERSONAL PROJECTS

ISITC — Academic Management System

- Developed an Angular 10 frontend implementing JWT-based user registration and login through `api/auth/signup` and `api/auth/signin` endpoints.
- Integrated the frontend with a Spring Boot backend and implemented JWT authentication handling with routing and an HTTP interceptor.

Technologies: Angular 10, TypeScript, Spring Boot, JWT, REST API

Brain MRI Image Classification

- Developed a deep learning solution for classifying brain MRI images for tumor detection using transfer learning and multiple pretrained CNN architectures.

Technologies: Python, TensorFlow, Keras, VGG16, ResNet50, EfficientNetB0, InceptionV3, OpenCV, Scikit-learn

Smarte-Procurement.ai

- Developed an intelligent tendering automation prototype that matches CVs with tender requirements, extracts skills, calculates matching scores, identifies missing/additional skills, and generates PDF evaluation reports.

Technologies: Python, NLTK, Scikit-learn, FPDF

Parkini — Subscription-Based Parking Platform

- Built a real-time parking management platform supporting simplified reservations and optimized subscription management.

Technologies: MongoDB, Express.js, React.js, Node.js

Messagerie — Real-Time Chat Application

- Developed a real-time one-to-one messaging application with live presence indicators and instant communication.

Technologies: Next.js, Node.js, Express.js, MongoDB, Socket.io

Chatbot Émotionnel — AI Chatbot

- Developed an AI-powered chatbot that generates absurd, philosophical, poetic, and humorous responses based on randomly selected personalities and conversation history.
- Integrated the Hugging Face Inference API with the Microsoft Phi-3-mini-4k-instruct language model for real-time response generation, with a fallback mechanism when the API is unavailable.
- Built a responsive chat interface with mood indicators, dark/light mode, animations, and interactive Easter eggs.

Technologies: Next.js 16, React 19, TypeScript, Tailwind CSS, Hugging Face Inference API, Microsoft Phi-3-mini-4k-instruct

TECHNICAL SKILLS

Programming Languages: Python, Java, JavaScript, PHP, HTML, CSS

Frameworks: Spring Boot, Symfony, React.js, Node.js, Express.js, Flutter

Databases: MySQL, PostgreSQL, MongoDB, PhpMyAdmin

Cloud & DevOps: AWS, Azure (basic knowledge), Docker, Kubernetes, Jenkins, Git

AI / Machine Learning: TensorFlow, CNN, NLP, ELK Stack

Other: UML, Figma, Power BI, gRPC, GraphQL, Kafka

LANGUAGES

- Arabic — Native
- French — Intermediate
- English — Intermediate
- German — Intermediate
- Spanish — Intermediate

CERTIFICATIONS

- IT Specialist – Databases
- CCNA1: Introduction to Networks
- CCNA2: Switching, Routing, and Wireless Essentials
- Cybersecurity Essentials

ASSOCIATIVE EXPERIENCE

Active Member — Google Club

- Regularly participated in workshops, technical events, and collaborative activities.

Member — Microsoft Club

- Took part in technology sessions and discussions around Microsoft solutions.

Member — CEP Entrepreneurship Club

- Participated in workshops, project ideation sessions, and entrepreneurship activities.

Hackathon “Nuit de l’Info” (2 editions)

- Contributed to the development of hackathon projects; served as team leader in one edition, overseeing team coordination, task distribution, and project organization.

PSC — Social and Cultural Project

- Renovated a classroom into a library at a local school and organized an educational event for children.